

LEDs' LIGHT INTENSITY CONTROLLER Based On PWM Technique

SHARAD BHOWMICK

This article describes how a simple circuit can be designed to control the intensity of a string of LEDs. The circuit is based on a common 555 timer IC

that employs pulse width modulation (PWM) technique to control the voltage. The 555 IC is driven in an astable mode. The same circuit can be used to control the speed of a DC

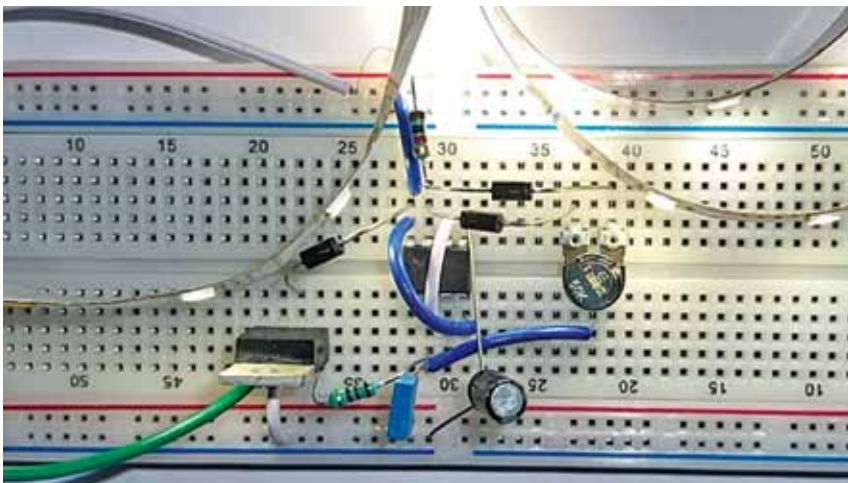
motor as well.

The most important component in this project is the 555 timer IC along with the MOSFET that works as a switch. The circuit works like a power electronics buck converter, which can be used to step down the DC voltage.

Components required for the project are listed in the table below.

BILL OF MATERIAL

- 555 Timer IC
- IRFZ44N
- 1k Resistor
- 200-ohm Resistor
- 100nF Capacitor
- 1nF Capacitor
- 3×1N4001 Diode
- 100k Potentiometer



The circuit on a breadboard

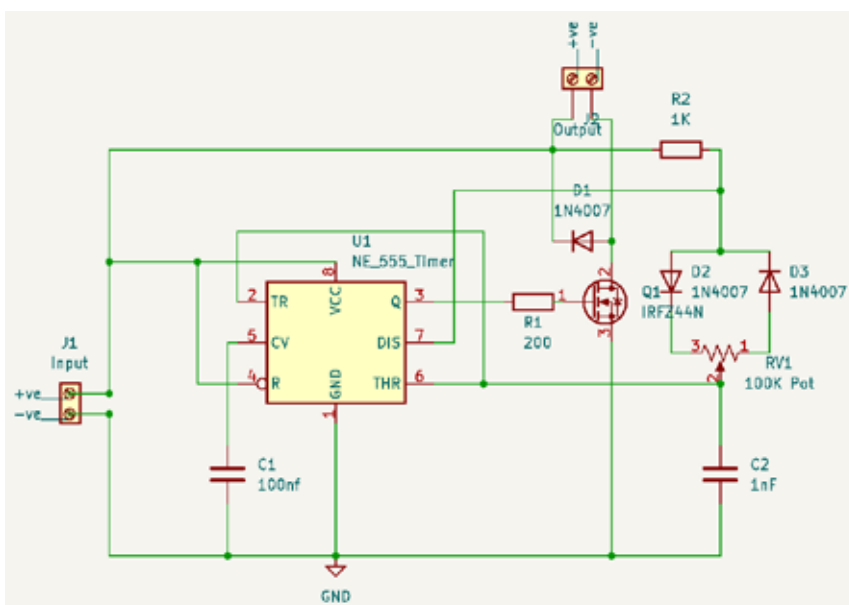


Fig. 1: Circuit diagram

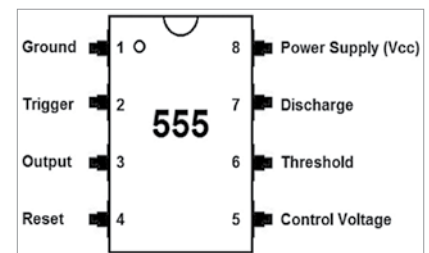


Fig. 2: Pin diagram of 555 timer

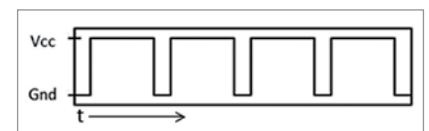


Fig. 3: Example of a PWM signal

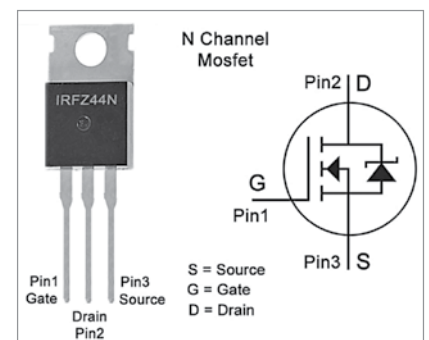


Fig. 4: Pin configuration of IRFZ44N